

Tushar Prajapati

AI Engineer & Full Stack Developer

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EXPERIENCE

IMARTICUS LEARNING | FULL STACK DEVELOPER (AI + NODE.JS + REACT)

Aug 2025 – Present | Noida

- **AI Pipeline (Gemini 2.5):** Designed a RabbitMQ batch-worker (priority queues, fault-tolerant retries) powering **AI Summarization**, **AI Tutor**, and **Grill Master** multi-turn mock-interview features — prompt engineering, function-calling, token-budget optimisation.
- **SPIPA LMS (Govt. of Gujarat):** Shipped an offline-first, LAN-deployed LMS for 200K+ users — 130ms avg. API, 93% Redis cache hit, 99.97% uptime over 30 days.
- **Security (VAPT):** Led PMRC (Govt. of India) VAPT audit resolution — closed 29 vulnerabilities incl. IDOR, JWT bypass, brute-force OTP, clickjacking, CSP hardening.
- **Infra:** Cut LLM token spend, automated HDFC bank reconciliation via AWS Lambda, and built the Hachiko health-check / alerting service for production.

KEY PROJECTS

SAIA --- ADAPTIVE AI AGENT (RAG + MEMORY + EVALS) | LLM APIs, VOYAGE

EMBEDDINGS, RAG, BULLMQ, MONGODB, REDIS

2026

- Built an autonomous conversational agent with a persistent **memory engine** (imprint / recall / salience), **RAG** (Voyage AI embeddings + cosine recall), **tool use / function calling**, and **multi-step planning** for proactive threads.
- Swappable LLM adapter (Gemini today, Anthropic / OpenAI drop-in) and an **eval harness** (identity / memory / continuity probes, ≥ 2 -of-3 gate) that runs against real LLMs before every release.

CLAUDE SKILLS LIBRARY --- 21-SKILL AGENT TOOLKIT | MCP-STYLE SKILL

PACKAGING, PROMPT ENGINEERING

2026

- Authored / curated 21 reusable agent **skills** (design taste, animation vocabulary, image-to-code, high-end visual design, etc.) packaged with a lock-file registry (`skills-lock.json`) that hashes and versions each skill — ships alongside a personal Claude Code / MCP setup.

CRUD & SUDOUSER --- MEAN + NEXT.JS PLATFORMS (RAVENEDGE) | REACT +

TYPESCRIPT, NODE, MONGODB, DOCKER, JWT

2026

- Two production platforms built in pnpm monorepos with TypeScript-strict everywhere: agile/CRM/campaigns for a team (**CRUD**) and a self-hosted single-user AI OS (**Sudouser**). React (App Router + hooks + Zustand-style state), JWT + argon2 auth, Docker Compose + Nginx.

ACHIEVEMENTS & PATENTS

- **Finalist** — Smart India Hackathon 2024 (Backend Lead)
- **Patents:** IoT-Enabled Street Light Regulation System (Jan 2024); Smart Field Personnel Monitoring System (Oct 2023)

SKILLS

LLM & AGENTS

OpenAI • Anthropic • Gemini • LangChain • LangGraph • LlamaIndex • CrewAI • AutoGen • MCP • RAG • Tool Use • Multi-step Planning • Evals

VECTOR DBS

pgvector • Chroma • Pinecone • Weaviate • Voyage

LANGUAGES & PYTHON WEB

Python • TypeScript • JavaScript • C++ • FastAPI • Flask • Django

FRONTEND

React (hooks, Redux) • Next.js • Tailwind • HTML5 / CSS3

BACKEND / APIS

Node.js • Express • REST • GraphQL • MongoDB • PostgreSQL • Redis

DEVOPS

Docker • Kubernetes • AWS (S3, EC2, Lambda) • CI/CD • Git

EDUCATION

AKGEC (AKTU)

B.TECH IN COMPUTER SCIENCE

2022 – 2026 | CGPA: 7.59 / 10

G.R. GLOBAL ACADEMY, DADRI

INTERMEDIATE

2021 | 94%

DSR MODERN SCHOOL, DADRI

HIGH SCHOOL

2019 | 95%

CERTIFICATIONS

Ethical Hacking & Countermeasures — Crow Security (2023)
EC-Council Cybersecurity for Business

CO-CURRICULAR

Co-founder, Centre for Future Studies and Research, AKGEC — led MARE'24 workshop and cybersecurity sessions.